Data Analysis Course – Fall 1405

Phase 1: Problem Understanding, Data Understanding, and Exploratory Analysis

Path A – NYC TLC Yellow Taxi Trip Records (January 2024)

Student: Aida Abbasi

Student ID: 401170897

Repository tag: phase-1

1. Problem Framing

NYC Yellow Taxi dispatch and fleet-positioning decisions are made ahead of demand: a dispatcher or a fleet-rebalancing system needs to know, before each hour starts, roughly how many pickups to expect in order to allocate cabs across the city. This project turns that operational need into an analyzable question: **can short-horizon citywide pickup demand be forecast from time-of-day, day-of-week, and recent history alone** — i.e. from information that is genuinely available before the hour in question, with no information from inside or after the forecast window.

The target output is a predicted trip count for a given (date, hour) cell, evaluated against the actual realized count for that cell. This is deliberately not a vague “best model” question: the output variable, time granularity, and evaluation window are fixed up front (Section 14).

2. Candidate Questions and Selected Main Question

Three candidate analytical questions were considered for this dataset:

- **Q1 (selected — time-series):** Can hourly citywide pickup demand be forecast from time-of-day, day-of-week and recent history, to support dispatch/supply planning?
- **Q2 (regression):** Can the fare amount of a trip be predicted from trip distance, pickup time and pickup/dropoff zone, using only information known before or at trip start?
- **Q3 (unsupervised):** Do pickup/dropoff zones cluster into interpretable demand “regimes” (e.g. commuter corridors vs. nightlife vs. airport) based on their hourly demand profile shape?

Q1 was selected as the main question. Justification: the course syllabus explicitly designates Path A (NYC TLC) as the project's time-series track, requiring a serious time-dependent analysis such as demand forecasting (see the “Why only one path?” note in the project brief). Q1 is also the most operationally direct framing of the dataset: trip-level fare prediction (Q2) and zone clustering (Q3) remain available as secondary/exploratory lines of analysis in later phases and are kept as the regression and unsupervised components of this same project, but Q1 is the question the whole pipeline (baseline, split, metrics) is built around.

3. Data Provenance

Source: NYC Taxi & Limousine Commission (TLC) Trip Record Data, official public dataset (*https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page*). File used: *yellow_tripdata_2024-01.parquet* (Yellow Taxi, January 2024), retrieved 2026-11-03 directly from the TLC's CloudFront distribution (*https://d37ci6vzurychx.cloudfront.net/trip-data/*). A companion static reference table, *taxi_zone_lookup.csv*, maps the integer location IDs to borough/zone names.

Unit of observation: one completed Yellow Taxi trip, as recorded by the vehicle's taxi meter and reported by the technology providers under TLC contract. Collection is a byproduct of the metering/payment system, not a designed survey — so values like distance and fare are meter-derived, while timestamps are logged automatically. The data is a full administrative census of metered Yellow Taxi trips for the month, not a sample.

Known limitations of the source itself: it covers only Yellow (street-hail) taxis — it excludes green boros taxis, for-hire vehicles (Uber/Lyft/Via), and all non-metered trips, so citywide ground-transportation demand is undercounted, especially outside Manhattan. Passenger counts are entered by the driver and are known in TLC documentation to be unreliable. Pickup/dropoff location is reported at the zone level (265 zones), not exact coordinates.

4. Structural Audit

Raw file: **2,964,624 rows × 19 columns**. Each row is one trip; there is no natural grouping/repetition within the file beyond that. The table below lists every column, its dtype, missingness and cardinality.

column	dtype	n_missing	pct_missing	n_unique
VendorID	int32	0	0.0	3
tpep_pickup_datetime	datetime64[us]	0	0.0	1575706
tpep_dropoff_datetime	datetime64[us]	0	0.0	1574780
passenger_count	float64	140162	4.728	10
trip_distance	float64	0	0.0	4489
RatecodeID	float64	140162	4.728	7
store_and_fwd_flag	object	140162	4.728	2
PULocationID	int32	0	0.0	260
DOLocationID	int32	0	0.0	261
payment_type	int64	0	0.0	5
fare_amount	float64	0	0.0	8970
extra	float64	0	0.0	48
mta_tax	float64	0	0.0	8
tip_amount	float64	0	0.0	4192
tolls_amount	float64	0	0.0	1127
improvement_surcharge	float64	0	0.0	5
total_amount	float64	0	0.0	19241
congestion_surcharge	float64	140162	4.728	6
Airport_fee	float64	140162	4.728	3

Relationship between files: *yellow_tripdata_2024-01.parquet* is a single flat table; *taxi_zone_lookup.csv* is joined to it via PULocationID/DOLocationID → LocationID (many trips to one zone).

5. Quality Audit

Each issue below is quantified against the raw row count, with the Phase 1 decision (kept / dropped / flagged) stated explicitly and justified.

issue	n_rows	pct_of_raw	initial_decision
Exact duplicate rows	0	0.0	Drop during cleaning (true duplicate, not a legitimate repeat trip)
Missing value in passenger_count	140162	4.728	Keep row; drop only in analyses specifically centered on passenger_count
Missing value in RatecodeID	140162	4.728	Keep row; RatecodeID is not used centrally in this phase
Missing value in congestion_surcharge / Airport_fee	140162	4.728	Keep row; these columns are not used in the main question (demand forecasting)
Pickup timestamp outside January 2024 (data-entry error)	18	0.001	Drop; outside the project's defined time scope
Implausible trip duration: zero or negative	870	0.029	Drop; timestamp error, physically impossible
Implausible trip duration: over 4 hours	1878	0.063	Drop; most likely a meter left running (abandoned trip), not a real ride
Implausible trip distance: zero or negative	60371	2.036	Drop; without distance, a real trip cannot be defined
Implausible base fare: zero or negative (fare_amount <= 0)	38341	1.293	Drop; likely a voided transaction or logging error
Implausible implied speed: over 80 mph	1107	0.037	Drop from the main analytic population; kept separately as an anomaly candidate (Section 12 of the report)
Zero passenger count (passenger_count == 0)	31465	1.061	Keep in the main analytic population (common and documented in TLC data); flag only

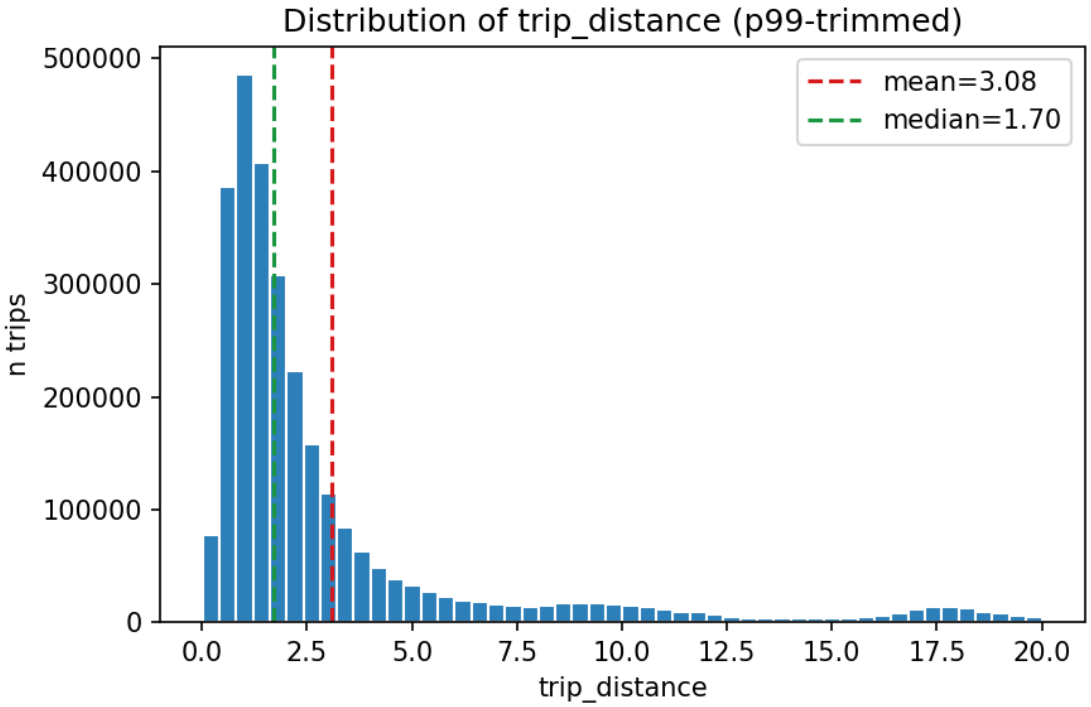
Cleaning funnel applied (ordered):

step	rows removed	rows remaining
excl_duplicate_rows	0	2,964,624
excl_outside_jan2024	18	2,964,606
excl_nonpositive_duration	870	2,963,736
excl_duration_over_4h	1,878	2,961,858
excl_nonpositive_distance	59,585	2,902,273
excl_nonpositive_fare	34,530	2,867,743
excl_speed_over_80mph	1,040	2,866,703

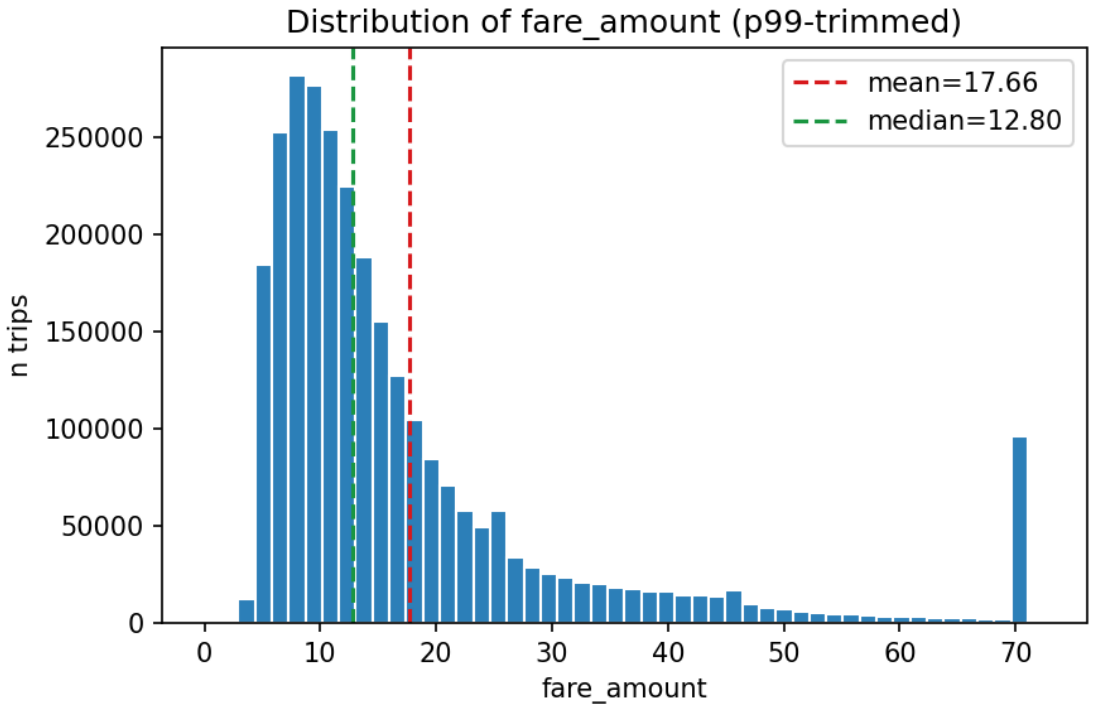
Final Phase-1 analytic population: 2,866,703 trips (96.7% of raw rows).

6. Univariate Exploratory Analysis

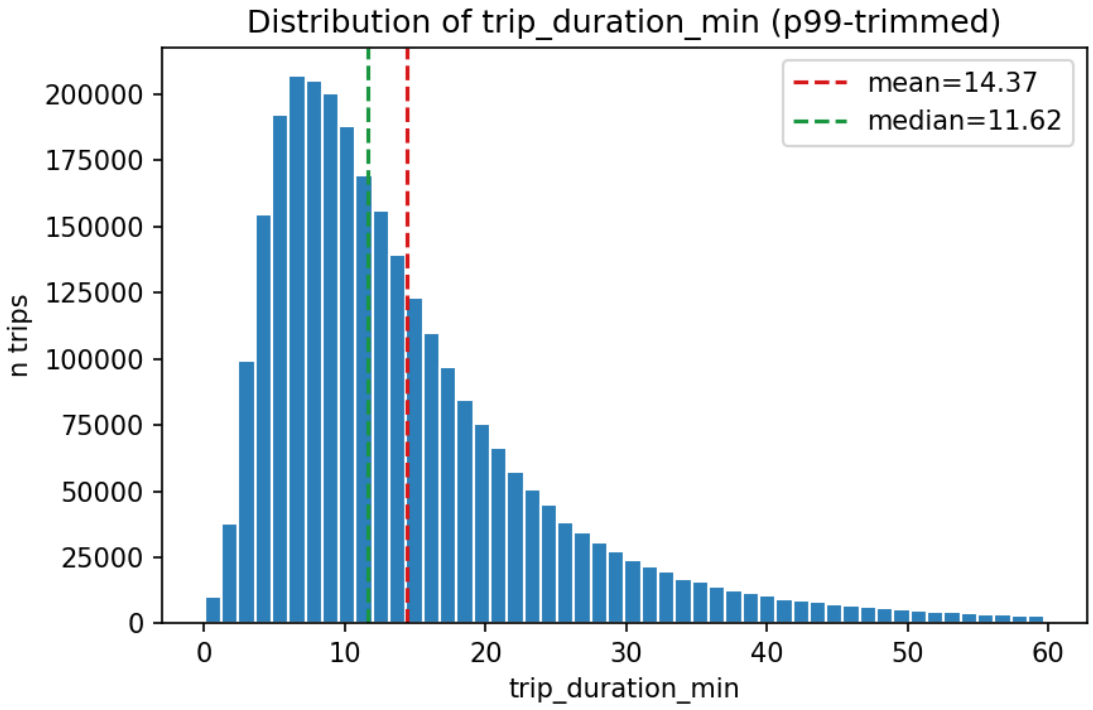
variable	mean	median	std	skew	min	p95	max
trip_distance	3.293	1.7	4.389	3.283	0.01	13.95	233.25
fare_amount	18.486	12.8	17.178	4.193	0.01	60.4	1616.5
total_amount	27.335	20.16	21.66	3.555	1.01	79.85	1617.5
trip_duration_min	14.968	11.717	11.864	2.575	0.017	37.883	238.717
speed_mph	11.516	9.672	6.719	2.046	0.005	26.155	80.0
passenger_count	1.339	1.0	0.851	3.045	0.0	3.0	9.0



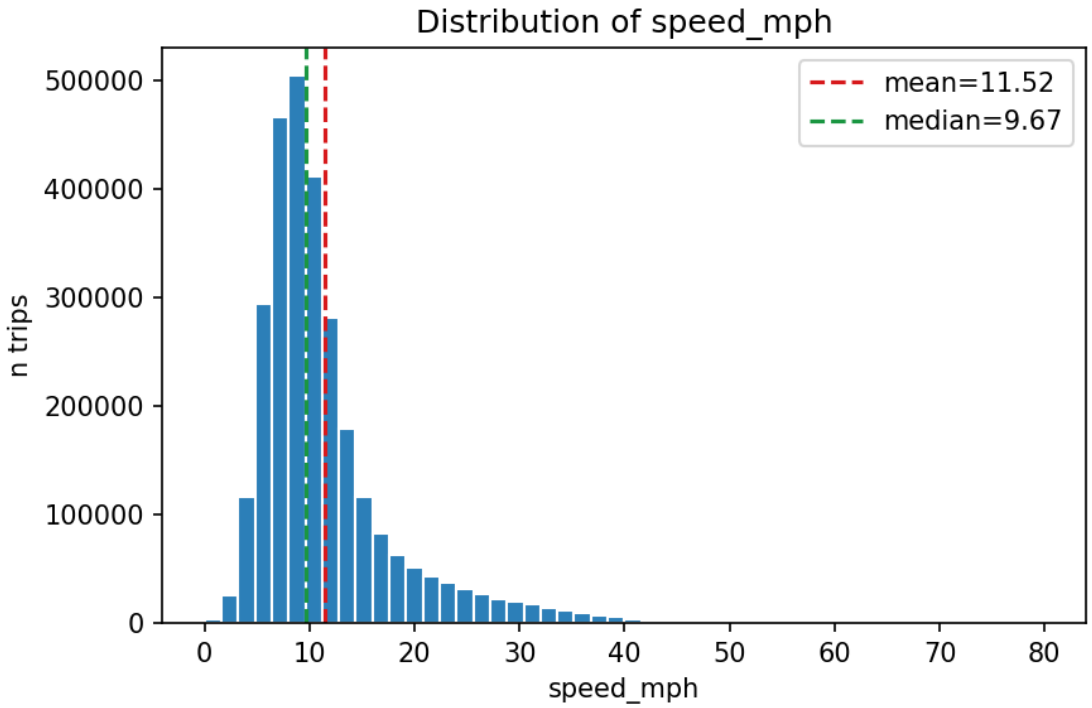
Trip distance is heavily right-skewed (median 1.7 mi, p95 14.0 mi) — typical of an urban taxi market dominated by short Manhattan hops.



Fare amount mirrors the distance skew; a long right tail reflects airport/long-haul trips.

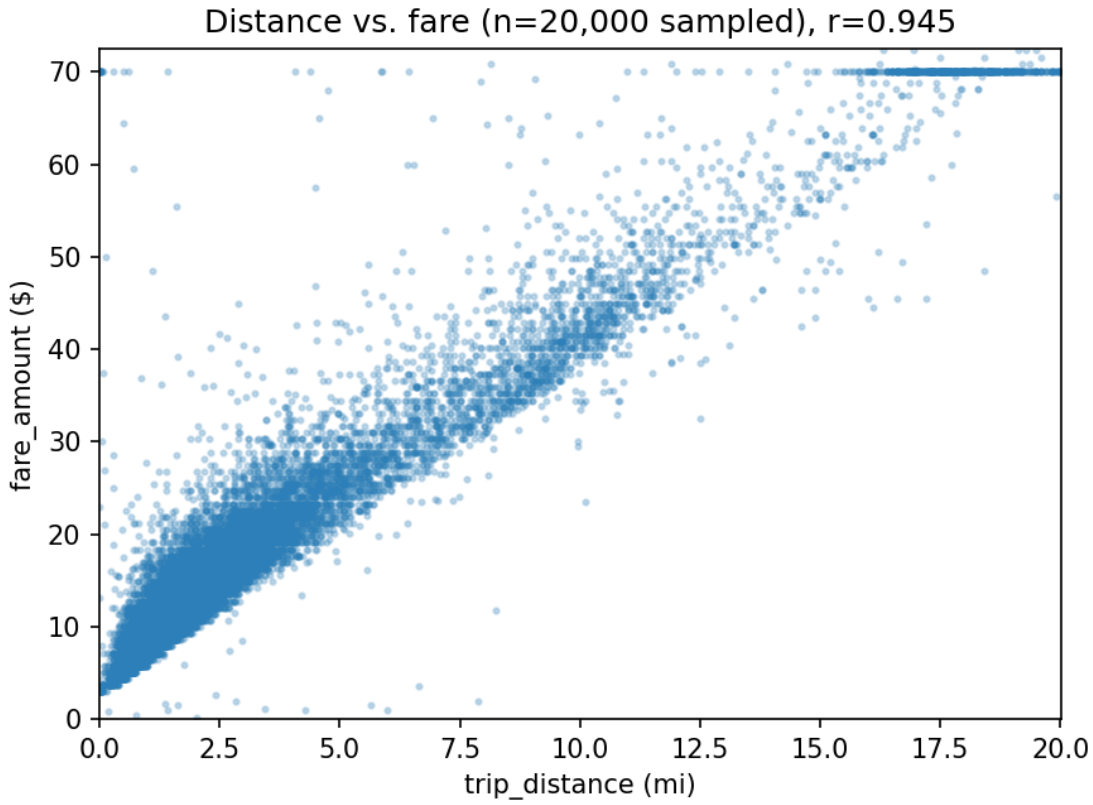


Trip duration is right-skewed around a ~12-minute median trip.

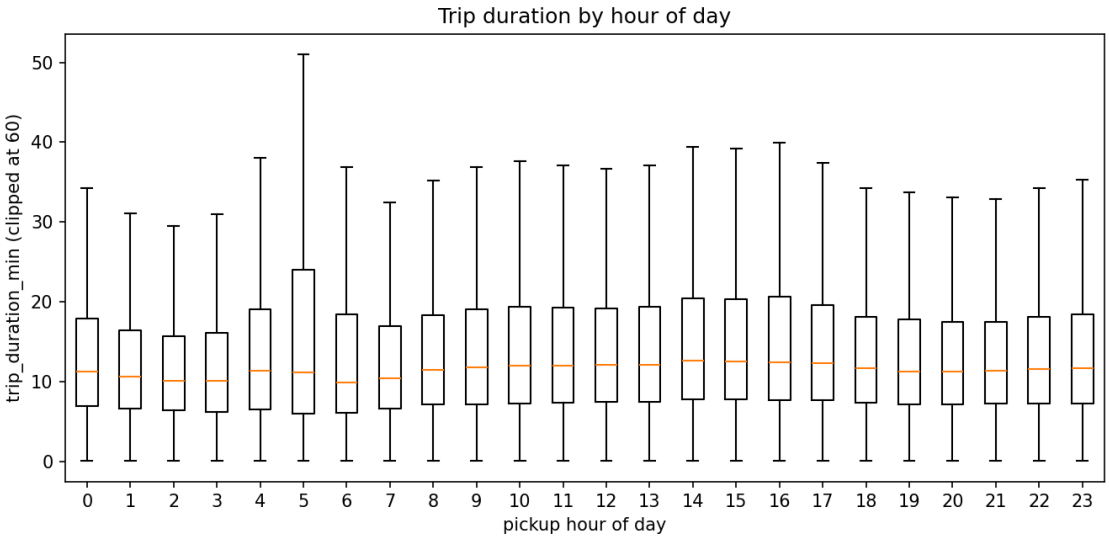


Implied speed (distance/duration) clusters around 8–12 mph, consistent with dense Manhattan traffic; the hard 80 mph cutoff used in cleaning is visible at the right edge.

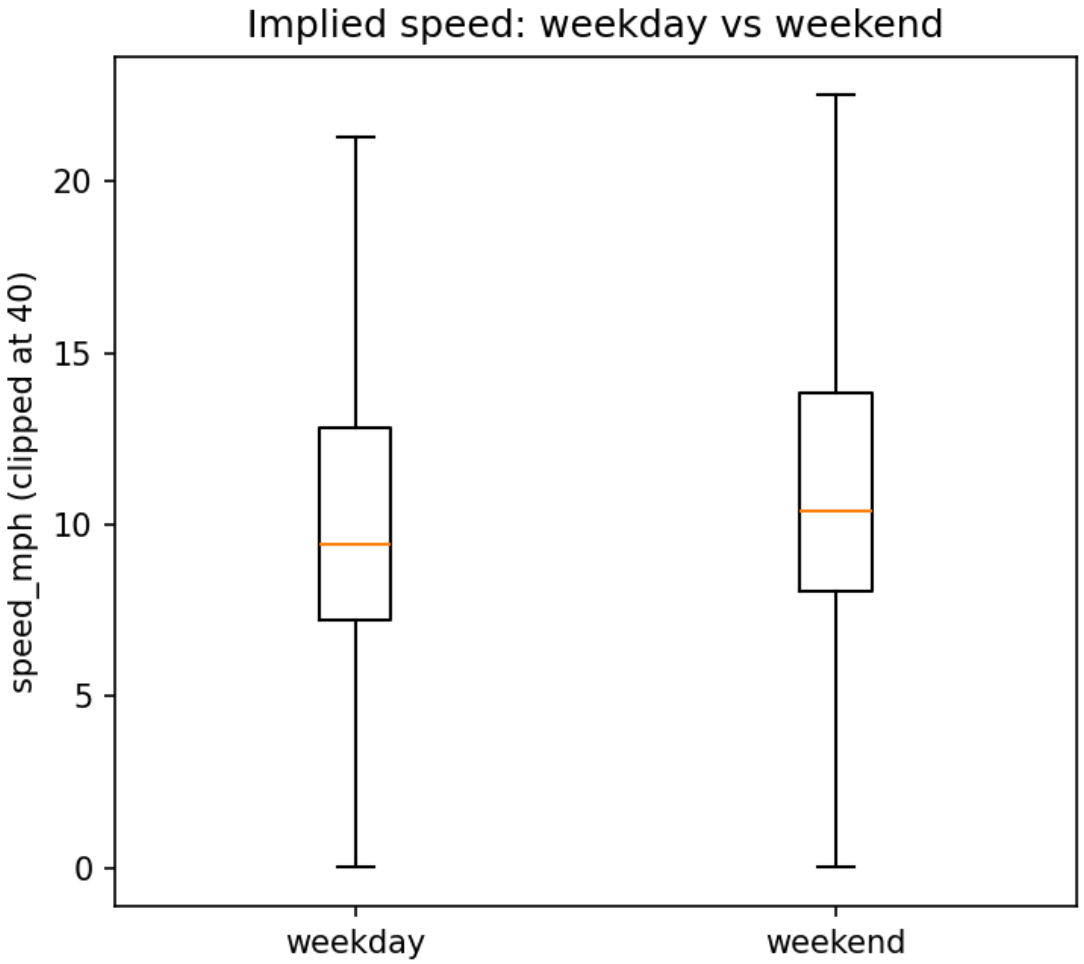
7. Multivariate Exploratory Analysis



Distance and fare are strongly, almost linearly related (as expected from the metered fare formula), which is exactly the relationship a Phase-3 regression baseline for fare prediction would need to beat.

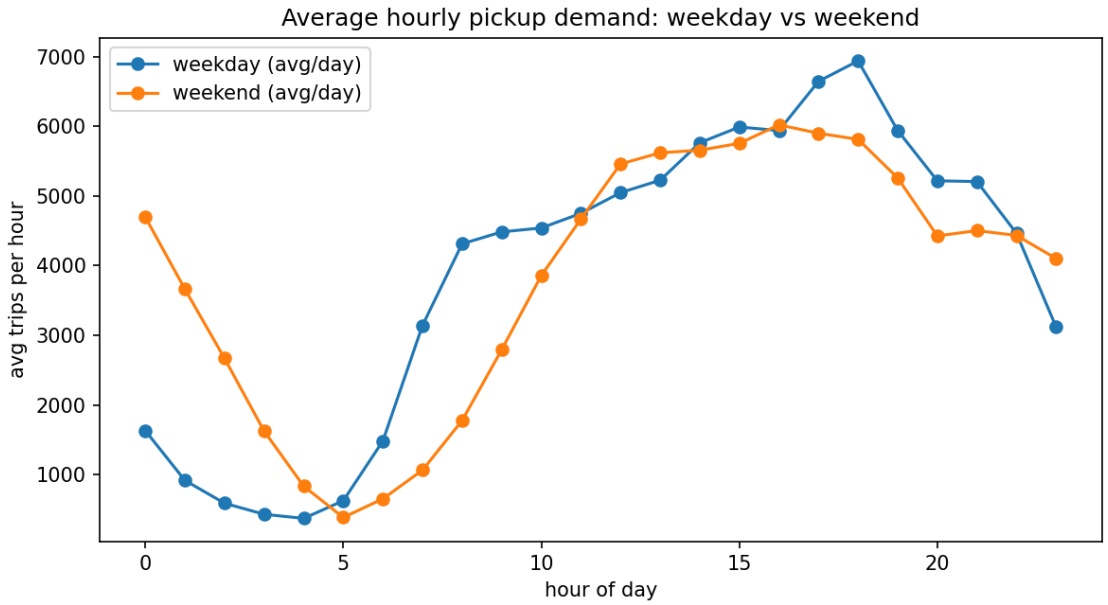


Trip duration for a given trip varies systematically with pickup hour — rush-hour pickups (8–9 and 16–18) take noticeably longer than overnight trips, evidence of a congestion effect that any duration/fare model should account for.

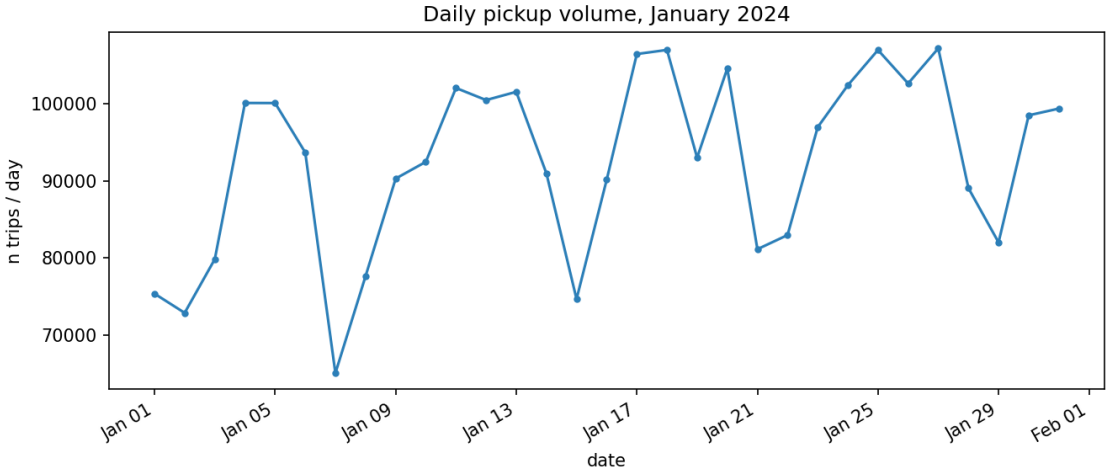


Implied speed is slightly higher on weekends than weekdays, consistent with lighter weekend road traffic.

8. Path A — Trip-Time Audit



Weekdays show a sharp bimodal commute pattern (peaks near hour 8 and a dominant peak at hour 18), while weekends show a flatter, later-starting profile with relatively higher demand in the very early morning (0–3h, nightlife traffic). This weekday/weekend shape difference is the core structure the Phase-3 forecasting model needs to capture.



Daily volume follows a clear 7-day cycle (lower on Mondays) with one sharp outlier: **January 8, 2024** drops to ~65k trips, far below the surrounding days (~90–100k). This date coincides with a winter storm that affected the NYC area in early January 2024; we note this as a plausible explanation, not a confirmed causal claim (Section 13), and flag it as a candidate weather-driven anomaly for the time-series model to either exclude or explicitly model in Phase 3.

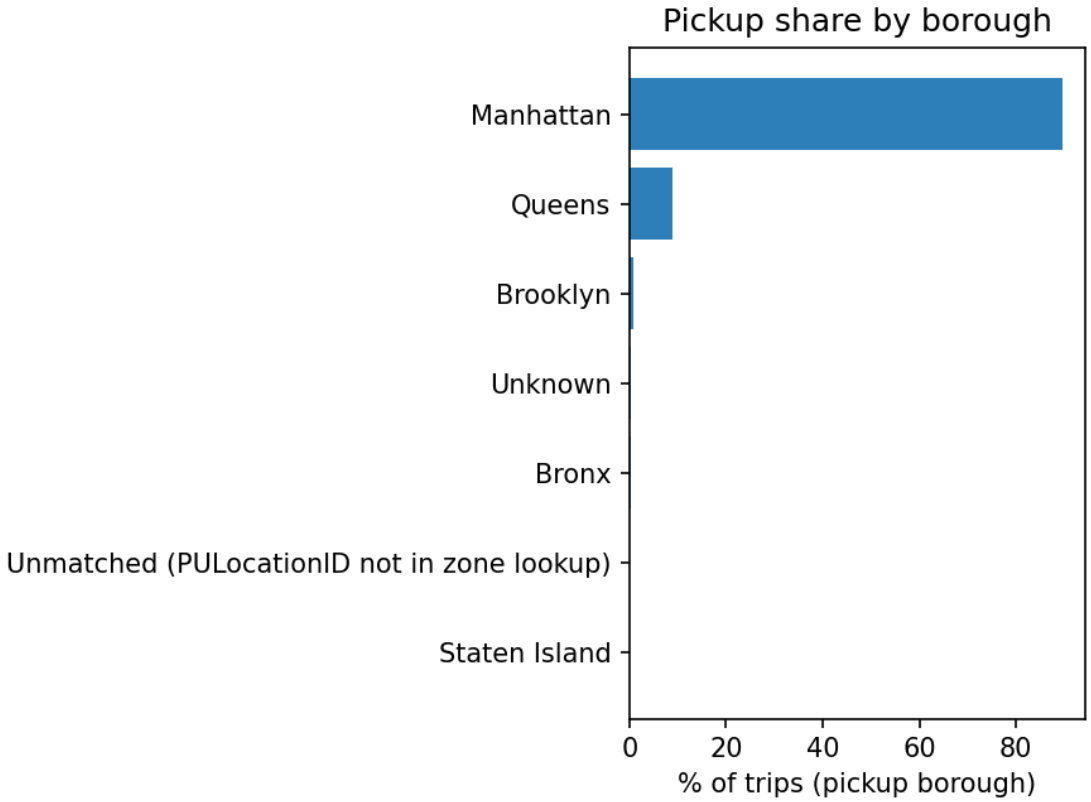
9. Path A — Variable Consistency

Distance, duration and fare are cross-checked via implied speed (speed_mph = trip_distance / (trip_duration_min/60)) and fare-per-mile (fare_amount / trip_distance). Implausible-speed records (>80 mph, n=1,040) were excluded from the main analytic population as data-entry errors; records with extreme fare-per-mile (top 0.1%, n=2,905) were kept but flagged as anomaly candidates (Section 12)

rather than dropped, since very short, high-surcharge trips (e.g. airport minimum fares) can legitimately produce a high $/mile ratio.

10. Path A — Initial Spatial Snapshot

borough	n_trips	pct_of_trips
Manhattan	2571245	89.69
Queens	257249	8.97
Brooklyn	22209	0.77
Unknown	9575	0.33
Bronx	5741	0.2
Unmatched (PULocationID not in zone lookup)	599	0.02
Staten Island	43	0.0
EWR	42	0.0



Pickups are extremely concentrated: Manhattan alone accounts for ~90% of all trips in the cleaned sample, Queens (largely JFK/LGA airport traffic) for ~9%, and the remaining boroughs together for <1%. This is a strong class/zone imbalance: any model or metric that reports only a citywide aggregate will be dominated by Manhattan behavior and could silently hide poor performance in the outer boroughs — a concrete instance of the zone-level evaluation requirement for this path.

Top 10 individual pickup zones:

zone	n_trips	pct
Upper East Side South	140044	4.89
Midtown Center	140044	4.89
JFK Airport	138161	4.82
Upper East Side North	133875	4.67
Midtown East	104277	3.64
Times Sq/Theatre District	102853	3.59
Penn Station/Madison Sq West	102077	3.56
Lincoln Square East	101690	3.55
LaGuardia Airport	87625	3.06
Upper West Side South	86412	3.01

11. Time-Series Question Definition

Question: forecast n_trips(date, hour) — the citywide pickup count for each hour — using only information available strictly before that hour begins (calendar features: hour-of-day, day-of-week, holiday flag; and lagged demand from prior hours/days). **Planned temporal split (Phase 3):** train on the first three weeks of January 2024 (Jan 1–21), validate on Jan 22–28, and hold out the final days of the month as test — chronological, with no shuffling, so no future hour is ever used to predict an earlier one.

12. Anomaly Candidates

anomaly_type	n_flagged	pct_of_raw
implausible_speed_gt_80mph	1107	0.0373
idle_meter_long_duration_zero_fare_move	9749	0.3288
very_high_fare_per_mile	2905	0.098
zero_passenger_count	31465	1.0613

These are retained in the data (not dropped) because each has a plausible benign explanation (idle meter while waiting for a passenger, a legitimate minimum/flat fare, an unreported passenger count) as well as a plausible error explanation; distinguishing the two needs more structure than Phase 1 EDA alone provides, so they are logged here as the starting point for Phase 4's error-taxonomy work.

13. Information Leakage Risk

- **total_amount, tip_amount, tolls_amount, improvement_surcharge, congestion_surcharge, Airport_fee** are all computed at or after trip completion (several are deterministic functions of fare_amount and the trip's characteristics). None of these may be used as predictors for a pre-trip fare or demand forecast — only post-hoc as the prediction target itself or for descriptive analysis.
- **tpep_dropoff_datetime** and any duration/speed feature derived from it leak directly into a duration- or arrival-time prediction task, since the dropoff time is only known once the trip is already over.
- For the selected **demand-forecasting** question specifically, the main leakage risk is using same-hour or future lagged-demand features (e.g. an aggregate that accidentally includes trips from the hour being predicted) — all lag features must be built from strictly past hours relative to the forecast hour.

14. Baseline Definition

Baseline for the main (demand-forecasting) question: predict next hour's trip count as the mean trip count observed in that same (day-of-week, hour-of-day) cell over the training weeks (a seasonal-naive baseline). It uses no model fitting, is fully interpretable, and must be beaten by any Phase-3 model for that model to be considered useful. **Secondary baseline for the fare-regression side question (Q2):** predict fare_amount as a simple linear function of trip_distance alone (median $/mile from the training data × distance), ignoring time, location and all other features.

15. Proposed Evaluation Metrics

- Demand forecasting (main question): MAE and RMSE of hourly trip counts; MAPE reported per zone/borough as well as citywide, to avoid Manhattan dominating the error average.
- Fare regression (secondary): RMSE and R^2 against the linear-distance baseline.

- Unsupervised zone clustering (secondary): qualitative interpretability of clusters plus silhouette score as a sanity check, not as the primary success criterion.

16. Preliminary Training / Validation / Test Split Plan

Chronological split only (see Section 11): train = Jan 1–21, validation = Jan 22–28, test = Jan 29–31. The same chronological boundaries will be reused for the fare-regression side question so that no trip after the training cutoff ever influences training in any analysis in this project.

17. Privacy, Ethics and Governance (preliminary)

The dataset contains no passenger names, payment-card numbers, or exact GPS coordinates — locations are reported at the 265-zone level specifically to limit re-identification risk, and this project does not attempt any join, enrichment or inference intended to re-identify a driver or passenger from the trip records. Pickup/dropoff zone and time, in combination, carry some residual re-identification risk for very rare origin-destination pairs at unusual hours (e.g. outer-borough late-night trips with only one or two monthly occurrences); any zone-level reporting in later phases will aggregate small cells rather than publish raw rare-pair counts.

18. Limitations and Next Steps

- Single month (January 2024, winter seasonality only) — findings about demand shape may not generalize to summer months; this is a scope limitation to be revisited if a multi-month extension is pursued.
- Only Yellow Taxi trips are covered; citywide ground-transportation demand (including For-Hire Vehicles) is necessarily undercounted.
- Passenger_count is self/driver-reported and known to be unreliable per TLC documentation; it is not used as a predictor anywhere in this project.
- Phase 2 will turn the cleaning/feature decisions above into a reproducible preprocessing pipeline and add the lag/calendar features needed for the Phase-3 forecasting model.